

Lab 5: Configuration of Static Router and Default Router

OBJECTIVES

- To understand the concept and importance of static routing in networks.
- To configure static routes on routers for specific network destinations.
- To verify connectivity between networks using static and default routing.
- To understand and configure a default route for unknown networks.

THEORY

Static Routing

Static routing is a method where network routes are manually configured on routers by the network administrator. Each route specifies a destination network, the subnet mask, and the next-hop address or exit interface. Static routes are simple, predictable, and do not change unless manually updated. They are efficient for small networks but can be difficult to manage in large or dynamic networks.

Advantages of Static Routing:

- Low overhead since routers do not exchange route information.
- More secure and predictable routing behavior.
- Useful for controlling traffic flow in a network.

Disadvantages:

- Not scalable for large networks.
- Requires manual updates if the network topology changes.

Default Routing

A default route is a special type of static route used to forward packets destined for networks not explicitly listed in the routing table. It acts as a “gateway of last resort.” The default route is typically used to direct traffic toward the Internet or an upstream router.

Example:

If a router has the following default route configured:

- `ip route 0.0.0.0 0.0.0.0 10.0.0.2`

All packets for unknown networks will be sent to the router at 10.0.0.2.

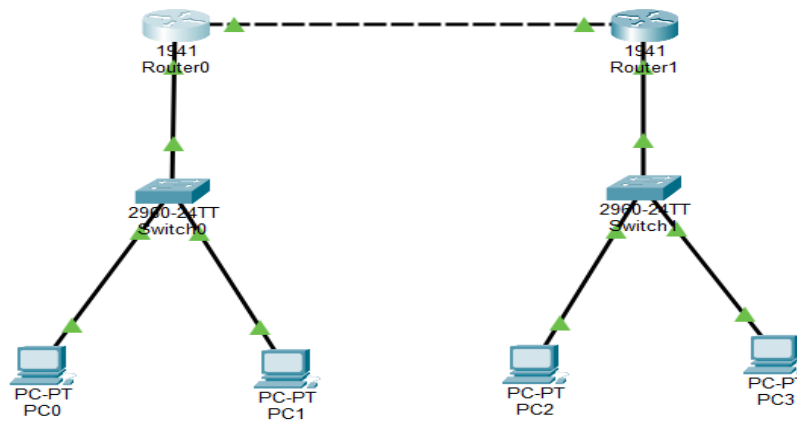
Verification:

After configuring static and default routes, the network administrator can verify the routing table using commands like show ip route and test connectivity using ping or traceroute. Correct configuration ensures proper packet delivery across different networks.

Configuration:

Device	Interface / Subnet	IP Address	Subnet Mask	Default Gateway
Router0	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A
Router0	FastEthernet0/1	10.0.0.1	255.0.0.0	N/A
Router1	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A
Router1	FastEthernet0/1	10.0.0.2	255.0.0.0	N/A
PC1	Subnet 1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	Subnet 1	192.168.1.3	255.255.255.0	192.168.1.1
PC3	Subnet 2	192.168.2.2	255.255.255.0	192.168.2.1
PC4	Subnet 2	192.168.2.3	255.255.255.0	192.168.2.1

Static Router



Router 0

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network 192.168.2.0

Mask 255.255.255.0

Next Hop 10.0.0.2

Add

Network Address

Router 1

Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network 192.168.1.0

Mask 255.255.255.0

Next Hop 10.0.0.1

Add

Network Address

Or we can also write command in CLI

Router 0

```
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2  
Router(config)#
```

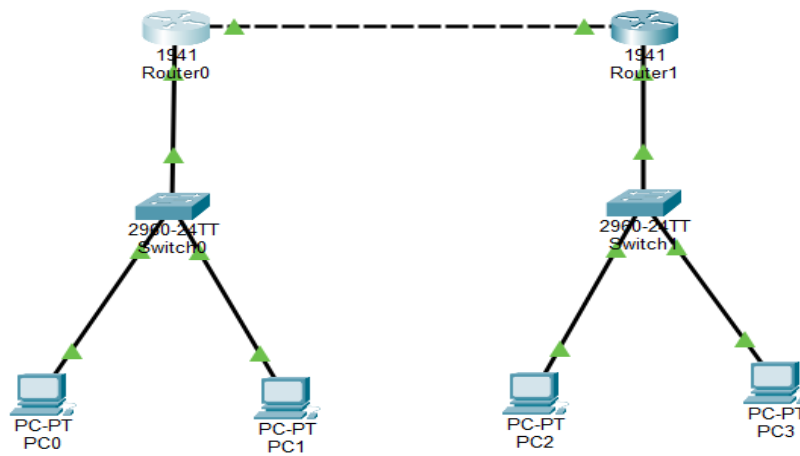
Router 1

```
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1  
Router(config)#
```

P0 Ping Output:

```
Pinging 192.168.2.2 with 32 bytes of data:  
  
Request timed out.  
Reply from 192.168.2.2: bytes=32 time=46ms TTL=126  
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126  
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126  
  
Ping statistics for 192.168.2.2:  
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 11ms, Maximum = 46ms, Average = 22ms
```

Default Router



Router 0

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network 0.0.0.0

Mask 0.0.0.0

Next Hop 10.0.0.2

Add

Network Address

0.0.0.0/0 via 10.0.0.2

Router 1

Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network 0.0.0.0

Mask 0.0.0.0

Next Hop 10.0.0.1

Add

Network Address

0.0.0.0/0 via 10.0.0.1

Or we can also write command in CLI

Router 0

```
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2  
Router(config)#
```

Router 1

```
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1  
Router(config)#
```

P1 Ping Output:

```
C:\>ping 192.168.2.3  
  
Pinging 192.168.2.3 with 32 bytes of data:  
  
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126  
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126  
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126  
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126  
  
Ping statistics for 192.168.2.3:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 10ms, Maximum = 10ms, Average = 10ms  
  
C:\>
```

RESULT

The configuration of static and default routing was successfully implemented and verified using the Cisco Packet Tracer environment. In the static routing setup, routes were manually configured on Router0 for the 192.168.2.0/24 network and on Router1 for the 192.168.1.0/24 network, which enabled end-to-end connectivity between the two subnets. Successful communication was confirmed through a ping test from PC0 to PC3 (192.168.2.2), where an initial 25% packet loss occurred due to ARP resolution, followed by stable replies with an average round-trip time of 22 ms. In the default routing configuration, specific static routes were replaced with a gateway of last resort (0.0.0.0 0.0.0.0), allowing both routers to forward traffic to unknown destinations. A ping test from PC1 to PC4 (192.168.2.3) resulted in 0% packet loss and a consistent round-trip time of 10 ms, confirming correct default route operation.

DISCUSSION

Static routing requires the network administrator to manually define routes in the routing table using the `ip route [destination_network] [subnet_mask] [next_hop]` command. This approach provides greater control, improved security, and low CPU overhead since routers do not exchange routing updates or perform dynamic path calculations. However, static routing lacks scalability and fault tolerance; any link failure would require manual reconfiguration to restore connectivity. Default routing, on the other hand, simplifies routing table entries by forwarding all unknown traffic to a predefined next hop using the `0.0.0.0 0.0.0.0` route. This method is especially effective in stub networks with a single exit path to external networks or the Internet. Network verification was carried out using the `ping` command, which confirmed successful data transmission. The initial request timeout observed during the first ping test is normal behavior and occurs while the router resolves the next-hop MAC address through the ARP process.

CONCLUSION

This lab successfully demonstrated the configuration and behavior of static and default routing between two distinct subnets. The experiment met its objective by enabling controlled packet forwarding and validating connectivity through appropriate testing methods. Static routing proved effective for small, stable networks requiring precise control, while default routing offered a simpler and more flexible solution for handling unknown destinations. Together, these routing techniques form essential foundations for efficient and predictable network communication.